

The Lorentz Transformation, Derived

Two sentences go in. One transformation comes out, and there is no second candidate. Everything else in this chapter is arithmetic.

WHERE WE ARE

Chapter 2.1 ended holding two postulates and a wreck. The wreck is the Galilean transformation $x' = x - vt, t' = t$: Maxwell's equations change form under it, the ether frame it demands cannot be found, and every attempt to save it — dragged ether, emission theory, contraction-by-fiat — died in the laboratory. The two postulates are what Einstein put in its place, and Chapter 2.1 deliberately stopped there.

This chapter asks the only question left: *what transformation between inertial coordinates could satisfy both postulates?* The answer is not "here is a clever guess that works." The answer is that once you also write down three structural assumptions that everyone makes silently — homogeneity, isotropy, reciprocity — the transformation is **forced**. There is one free function of v in the general form, and the second postulate plus reciprocity pin it down completely. Nobody is choosing anything.

Then we spend it. Simultaneity, time dilation and length contraction are corollaries, derived in that order and for a reason: the second and third are *consequences* of the first, and the standard presentation, which does them backwards, is why so many people never recover. After that: how velocities combine, why they seem not to add, and the variable in which they do. The chapter closes by taking the two famous paradoxes apart and showing that they are the same paradox wearing different clothes.

One warning about what this chapter is *not*. It is not geometry. We will produce an invariant combination $c^2t^2 - x^2$ almost by accident in §2 and then mostly leave it alone. Turning that invariant into a genuine geometry — a spacetime with a notion of length, angle and straightness — is Chapter 2.3's job, and it is a better chapter for having this algebra already in hand.

TOOLS YOU'LL NEED — Chapter 2.1 §7: the two postulates, which are the entire input. Chapter 0.4: linear maps, matrices, matrix multiplication, and what it means for a set of maps to be closed under composition — §3 is that, applied. Chapter 0.3: Taylor and binomial expansions, used every time we ask "and what does this look like at ordinary speeds?" Chapter 0.1 §3: the derivative as the coefficient of a linear approximation — the linearity argument in §1 is that idea run in reverse, and §5 differentiates a coordinate transformation.

1 · What the postulates constrain

Start by writing down exactly what we are given, and then — this is the part usually skipped — exactly what we are quietly assuming on top of it.

1.1 · The input

THE TWO POSTULATES

P1 (principle of relativity). The laws of physics take the same form in all inertial frames. No experiment inside a uniformly moving laboratory can reveal its velocity.

P2 (invariance of c). Light propagates in vacuum with speed c , the same in all inertial frames, independent of the motion of the source.

Fix the geometry once and use it for the whole chapter. Two inertial frames S and S' , axes parallel, S' moving in the $+x$ direction of S with speed v , and the origins coinciding at the moment both clocks read zero. This is the **standard configuration**, and it costs nothing: any other arrangement is this one composed with a rotation and a shift of origin. As throughout Part II, write the relative speed in units of c :

$$\beta \equiv \frac{v}{c}. \quad (2.2.1)$$

Nothing is assumed yet about the range of β ; that $|\beta| < 1$ will emerge in §2.3, where a larger value makes the transformation complex, and be proved unreachable in §5.

We want the four functions that take an event's coordinates (t, x, y, z) in S to its coordinates (t', x', y', z') in S' . An *event* is a point of spacetime — a flash, a collision, a clock tick — and both observers are describing the same one. Nothing physical changes when we relabel; the question is only what the labels are.

1.2 · The three assumptions nobody states

The postulates alone do not determine the transformation. They are constraints on a class of maps, and you have to say which class. Three assumptions do that work, and it is worth being blunt: **these are physical assumptions, not logical necessities**. Each is testable, each has been tested, and each could in principle have come out otherwise. This is where the real content of the chapter enters, and pretending otherwise is how the derivation acquires its undeserved reputation for being a magic trick.

(H) Homogeneity of space and time

No place is special and no moment is special. Concretely: the rule relating S -coordinates to S' -coordinates is the same rule everywhere and everywhen. If you and I are both inertial observers, the dictionary between our coordinates cannot depend on where in the room we set up, or on what year it is.

This forces the transformation to be linear. Here is the argument, and it is better than the hand-wave usually offered.

Let Φ be the map from S -coordinates to S' -coordinates, acting on events. Take any two events P and Q and let $\Delta = Q - P$ be their coordinate separation in S . Homogeneity says the separation the primed observer assigns, $\Phi(Q) - \Phi(P)$, can depend on Δ but *not* on where the pair sits — otherwise a rod's transformed length would tell you your absolute position, and space would have a marked spot. So there is a single function F with

$$\Phi(P + \Delta) - \Phi(P) = F(\Delta) \quad \text{for every event } P. \quad (2.2.2)$$

Now chain two separations. Go from P to $P + \Delta_1$ to $P + \Delta_1 + \Delta_2$ and add the two pieces:

$$\begin{aligned} F(\Delta_1 + \Delta_2) &= \Phi(P + \Delta_1 + \Delta_2) - \Phi(P) \\ &= [\Phi(P + \Delta_1 + \Delta_2) - \Phi(P + \Delta_1)] + [\Phi(P + \Delta_1) - \Phi(P)] \\ &= F(\Delta_2) + F(\Delta_1). \end{aligned} \quad (2.2.3)$$

That is **Cauchy's functional equation**, and for a continuous F its only solutions are the linear ones, $F(\Delta) = M\Delta$ for a constant matrix M . Since the origins coincide, $\Phi(0) = 0$ and therefore $\Phi(\Delta) = F(\Delta) = M\Delta$: the transformation is a constant matrix acting on (ct, x, y, z) . The entries of M may depend on v — they had better — but not on the event.

▼ Grind box — Cauchy's equation, and the loophole that "straight lines stay straight" leaves open

Cauchy. ▀ The theorem is: if $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ satisfies $F(a + b) = F(a) + F(b)$ and is continuous at even one point, then F is linear. The sketch in one dimension: additivity gives $F(na) = nF(a)$ for integer n , hence $F(a) = F(n \cdot a/n) = nF(a/n)$, so $F(qa) = qF(a)$ for every rational q ; continuity then upgrades rational to real. (Without continuity the theorem is false — there exist monstrous non-measurable additive functions, built with the axiom of choice. Nobody is proposing that spacetime is coordinatised by one.)

The other argument, and why it is not quite enough. The usual alternative is: a free particle moves uniformly in a straight line in S (Newton's first law), and by P1 it must do so in S' too, so the map takes straight worldlines to straight worldlines — therefore it is linear. The conclusion is right but the reasoning has a gap. The maps of $\mathbb{R}^n$ carrying straight lines to straight lines are not just the affine ones; they are the **projective** ones,

$$x'^\mu = \frac{A^\mu_\nu x^\nu + b^\mu}{c_\nu x^\nu + d},$$

which are perfectly good line-preserving maps with a nonconstant denominator. ▀ (That every line-preserving bijection is of this form is the fundamental theorem of projective geometry; we quote it.) Such a map is not linear, and it does something revealing: it blows up on the hypersurface $c_\nu x^\nu + d = 0$. That is a preferred set of events — a marked place and time in the universe. **Homogeneity is exactly the assumption that kills the denominator**, forcing $c_\nu = 0$ and returning us to affine, hence (with coinciding origins) linear.

This is not an idle technicality. Theories that keep P1 and P2 but drop strict homogeneity — "de Sitter relativity", the Fock–Lorentz transformations — live precisely in that discarded denominator, and they are constrained by experiment rather than by logic. The assumption is doing work, and now you know which work.

(I) Isotropy of space

No direction is special either. Two consequences, both needed.

First: $y' = y$ and $z' = z$. Directions perpendicular to the boost are untouched. Suppose instead $y' = k(\beta)y$ for some factor k . Boosting by $-v$ afterwards must return us to S , so $k(\beta)k(-\beta) = 1$. But isotropy says the transverse factor cannot know the *sign* of the boost — reflect the x -axis and a $+v$ boost becomes a $-v$ boost while y is untouched — so $k(\beta) = k(-\beta)$. Together: $k^2 = 1$, and since $k(0) = 1$ and k is continuous, $k = 1$.

Second: t' and x' cannot depend on y or z . By linearity, $t' = Dt + Ex + Gy + Hz$. Rotate the whole apparatus by 180° about the x -axis. This sends $y \rightarrow -y$, $z \rightarrow -z$ and leaves the boost configuration exactly as it was, so it cannot change t' — which forces $G = H = 0$. Same argument for x' .

▼ Grind box — the transverse argument done physically, with rods and scratches

The algebraic argument above is airtight but feels like bookkeeping. Here is the physical version, which is the one to remember, because it shows that a transverse contraction would produce a flat *contradiction* rather than merely a strange world.

Two identical rods, each of proper length one metre, both aligned along y , with their lower ends riding along the x -axis. One is at rest in S , the other at rest in S' , and they slide past each other. Fix a sharp scribe at the top of each rod.

Suppose S finds the moving rod short, $k < 1$. Then as they pass, the S -rod's scribe passes *above* the tip of the S' -rod, and the S' -rod's scribe scratches the S -rod's side below its tip. Whether a scratch appears on a rod is a local coincidence of matter with matter: it is a single event, and every observer agrees it happened.

Now look from S' . By P1, S' 's situation is the mirror of S 's — S' sees S moving at $-v$, and by isotropy the physics cannot distinguish $+v$ from $-v$ here. So S' must equally find *the other* rod short, and predicts the scratches the other way round. Both frames cannot be right about which rod carries a scratch, because that is one event and not two. The only escape is $k = 1$: the tips meet exactly, nobody scratches anybody, and both frames agree.

Note carefully what did *not* happen: no such contradiction arises along x . There, "which rod is shorter" is settled by measurements at two separate places, and §4.3 will show that two separate places is precisely where frames stop agreeing about what "at the same time" means. Transverse: one event, no room to disagree. Longitudinal: two events, all the room in the world.

(R) Reciprocity

If S' moves at $+v$ as measured in S , then S moves at $-v$ as measured in S' . Equivalently: the transformation from S' back to S is the same transformation with $v \rightarrow -v$.

This sounds like a tautology and it is not. It is the statement that the two frames stand in a symmetric relation — that neither is privileged in a way that would let S measure S' receding at one rate while S' measures S receding at another. It follows from P1 together with isotropy, and we could derive it; we will instead assume it and flag it, because assuming it makes visible exactly how much is being asked.

⚠ WHY THIS ISN'T OBVIOUS

Notice what has and has not been assumed. We have *not* assumed $t' = t$. We have not assumed that lengths are invariant, that simultaneity is absolute, that velocities add, or that there is a universal "now". Every one of those was baked into the Galilean transformation without comment for three hundred years, and every one of them is about to fail.

What we *have* assumed is that spacetime is uniform and directionless and that the relation between inertial observers is symmetric. Those three, plus the two postulates, are the complete input. If the answer surprises you, the surprise has to be located in one of those five statements — there is nowhere else for it to hide. That is the point of listing them.

2 · The derivation, in full

By §1 the transformation is linear, transverse coordinates are untouched, and nothing depends on y or z . So the whole problem lives in the (t, x) plane and reads

$$x' = Ax + Bt, \quad t' = Dt + Ex, \quad y' = y, \quad z' = z, \quad (2.2.4)$$

where A, B, D, E are constants — functions of v alone. Four unknowns. We have three conditions, and the fourth is fixed by the fact that an overall rescaling of both frames is not a physical transformation. Let us impose them one at a time.

2.1 · Condition (i): the origin of S' moves at v

This is what " S' moves at v " means, made operational. The spatial origin of S' is the locus $x' = 0$, and by construction it travels along $x = vt$ in S . Substituting into (2.2.4):

$$0 = A(vt) + Bt \quad \text{for all } t \quad \implies \quad B = -Av. \quad (2.2.5)$$

So the spatial part is already forced into the familiar shape

$$x' = A(x - vt). \quad (2.2.6)$$

Notice how little that cost, and notice what it did *not* do: it says nothing whatsoever about t' . The Galilean transformation is the special case $A = 1, D = 1, E = 0$. Every departure from Galileo is now compressed into three numbers.

2.2 · Condition (ii): a light pulse goes at c in both frames

This is Postulate 2, and it is the crucial input — the only place where the physics of light enters. At the moment the origins coincide ($t = t' = 0, x = x' = 0$) a flash is emitted from the common origin. Consider the part of it travelling in the $+x$ direction. In S it obeys $x = ct$. In S' , by P2, it must obey $x' = ct'$ — the *same* c , not $c - v$.

Put $x = ct$ into (2.2.6) and into $t' = Dt + Ex$:

$$x' = A(ct - vt) = At(c - v), \quad t' = Dt + E(ct) = t(D + Ec), \quad (2.2.7)$$

and demand $x' = ct'$:

$$A(c - v) = c(D + Ec) = cD + Ec^2. \quad (2.2.8)$$

Now do the same for the part of the flash going in the $-x$ direction, $x = -ct$, which must obey $x' = -ct'$:

$$-A(c + v) = -c(D - Ec) \implies A(c + v) = cD - Ec^2. \quad (2.2.9)$$

Two linear equations, two unknowns D and E in terms of A . Add (2.2.8) and (2.2.9): the Ec^2 terms cancel and

$$2Ac = 2cD \implies D = A. \quad (2.2.10)$$

Subtract them: the cD terms cancel and

$$-2Av = 2Ec^2 \implies E = -\frac{Av}{c^2}. \quad (2.2.11)$$

Three of the four unknowns are gone, and everything is expressed through A :

$$x' = A(x - vt), \quad t' = A\left(t - \frac{vx}{c^2}\right). \quad (2.2.12)$$

Pause on (2.2.12), because two things have already happened that no amount of staring at Galileo would have suggested.

The time equation has picked up a position term. t' depends on x . Two events at the same t but different x have different t' . Simultaneity has already broken, before we know what A is, and it broke because of the second postulate and nothing else. Everything in §4 is contained in the term $-Avx/c^2$.

The same coefficient A appears in both. That was not imposed; it fell out of adding the two light conditions. The transformation is symmetric between ct and x in a way that is easier to see if we measure time in metres — that is, use $x^0 = ct$ as the coordinate rather than t , as Part II does throughout. Multiplying the second equation by c :

$$ct' = A(ct - \beta x), \quad x' = A(x - \beta ct). \quad (2.2.13)$$

Now ct and x enter on exactly the same footing. That symmetry is the first hint that time is about to become a coordinate like the others, which is the whole thesis of Chapter 2.3.

2.3 · Condition (iii): reciprocity fixes A

One unknown left. Reciprocity says the inverse of the boost by $+v$ is the boost by $-v$. Apply the boost with parameter β and then the boost with parameter $-\beta$, and demand that we get back what we started with. Writing $A_+ = A(\beta)$ and $A_- = A(-\beta)$:

$$\begin{aligned} x'' &= A_-(x' + \beta ct') \\ &= A_- [A_+(x - \beta ct) + \beta A_+(ct - \beta x)] \\ &= A_- A_+ [x - \beta ct + \beta ct - \beta^2 x] = A_- A_+ (1 - \beta^2) x. \end{aligned} \quad (2.2.14)$$

For this to equal x for every x we need

$$A(\beta) A(-\beta) (1 - \beta^2) = 1. \quad (2.2.15)$$

(The same manipulation on ct'' gives the same condition; see the grind box.) One equation, two unknowns — unless A is an even function of β , which is exactly what isotropy delivers. Reflect the x -axis: set $X = -x$, $X' = -x'$. Then (2.2.13) becomes

$$ct' = A(\beta)(ct - (-\beta)X), \quad X' = A(\beta)(X - (-\beta)ct), \quad (2.2.16)$$

which is a boost with parameter $-\beta$ and coefficient $A(\beta)$. Space has no handedness, so the reflected description is as good as the original, and the coefficient for parameter $-\beta$ must be what it always is: $A(-\beta) = A(\beta)$. Feed that into (2.2.15):

$$A^2(1 - \beta^2) = 1 \quad \implies \quad A = \pm \frac{1}{\sqrt{1 - \beta^2}}. \quad (2.2.17)$$

Take the positive root: $A(0) = +1$ (no boost is no change), A is continuous, and A never vanishes, so it cannot cross zero to reach the negative branch. Name it:

$$\gamma \equiv \frac{1}{\sqrt{1 - \beta^2}} = (1 - \beta^2)^{-1/2}. \quad (2.2.18)$$

And there it is.

$$\boxed{ct' = \gamma(ct - \beta x), \quad x' = \gamma(x - \beta ct), \quad y' = y, \quad z' = z.} \quad (2.2.19)$$

That is the **Lorentz transformation** for a boost along x , in the form this book uses everywhere. Note that $\gamma \geq 1$ always, that $\gamma \rightarrow \infty$ as $\beta \rightarrow 1$, and that $\beta > 1$ would make γ imaginary — a first, cheap indication that c is a barrier, made respectable in §5.

▼ Grind box — the inverse, checked both ways, and the time equation of the reciprocity step

The time half of (2.2.14). Same manoeuvre:

$$\begin{aligned} ct'' &= A_-(ct' + \beta x') = A_- A_+ [(ct - \beta x) + \beta(x - \beta ct)] \\ &= A_- A_+ (1 - \beta^2) ct, \end{aligned}$$

which gives condition (2.2.15) again. Consistent, as it must be — two equations, one condition.

Inverting (2.2.19) directly. Do not appeal to reciprocity; just solve. Multiply the first equation by γ and the second by $\gamma\beta$ and add:

$$\gamma ct' + \gamma\beta x' = \gamma^2 [(ct - \beta x) + \beta(x - \beta ct)] = \gamma^2 (1 - \beta^2) ct = ct,$$

since $\gamma^2 (1 - \beta^2) = 1$ is (2.2.18) squared and rearranged. The same trick with the roles swapped gives x . Hence

$$ct = \gamma(ct' + \beta x'), \quad x = \gamma(x' + \beta ct'),$$

which is (2.2.19) with $\beta \rightarrow -\beta$ and primes swapped. So reciprocity, which we assumed, is *also* a theorem of the result — a genuine consistency check rather than a circularity, because the algebra above never used it.

Numbers worth memorising. γ is stubbornly close to 1 until β is large:

β	0.01	0.1	0.5	0.8	0.9	0.99	0.999
γ	1.00005	1.005	1.155	1.667	2.294	7.089	22.37

Half the speed of light buys you a 15% effect. This is exactly why nineteenth-century physics could be built without noticing: at $\beta = 10^{-4}$, the Earth's orbital speed, $\gamma - 1 = 5 \times 10^{-9}$.

2.4 · A consistency check we did not pay for

Postulate 2 was imposed only along the x -axis — one pulse forward, one back. But light goes in all directions, and a flash from the origin makes an expanding sphere, $x^2 + y^2 + z^2 = c^2 t^2$ in S . If the derivation is any good, that sphere had better still be a sphere in S' , expanding at the same c . We never asked for that. Check it.

Compute the combination $c^2 t'^2 - x'^2$ from (2.2.19):

$$\begin{aligned} c^2 t'^2 - x'^2 &= \gamma^2 (ct - \beta x)^2 - \gamma^2 (x - \beta ct)^2 \\ &= \gamma^2 [c^2 t^2 - 2\beta ctx + \beta^2 x^2 - x^2 + 2\beta ctx - \beta^2 c^2 t^2] \\ &= \gamma^2 (1 - \beta^2) (c^2 t^2 - x^2) = c^2 t^2 - x^2. \end{aligned} \tag{2.2.20}$$

The cross terms cancel identically and $\gamma^2(1 - \beta^2) = 1$ finishes it. Since $y' = y$ and $z' = z$, we have proved something considerably stronger than we asked for:

$$c^2 t'^2 - x'^2 - y'^2 - z'^2 = c^2 t^2 - x^2 - y^2 - z^2. \tag{2.2.21}$$

So the set $c^2 t^2 - x^2 - y^2 - z^2 = 0$ — the expanding light sphere — maps to itself. Light emitted from the common origin is a sphere of radius ct in S and a sphere of radius ct' in S' , centred on *each observer's own origin*, which are moving apart. Both are right. If that sounds impossible, hold the thought until §4.1; the two observers disagree about which points of the sphere are "simultaneous", and that disagreement is exactly large enough.

THE QUANTITY THAT SURVIVED

(2.2.21) was not an input. We demanded invariance of the light speed — a statement about one particular family of events, the ones with $c^2 t^2 - x^2 - y^2 - z^2 = 0$ — and got back invariance of the whole quantity, for *every* pair of events, whatever its value. That is an enormous strengthening, and it is the single most important line in this chapter.

The combination $\Delta s^2 = c^2 \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2$ is called the **interval**. Everybody computes the same number for it. It plays the role that $\Delta x^2 + \Delta y^2 + \Delta z^2$ plays for rotations in ordinary space — the thing that does not care how you turned your head. Chapter 2.3 takes that analogy seriously and builds a geometry out of it, and Chapter 2.4 turns the invariance into the algebraic condition $\eta_{\mu\nu} \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma = \eta_{\rho\sigma}$ that defines the Lorentz group. Everything downstream of here rests on one minus sign.

2.5 · The low-speed limit: Galileo was not wrong, he was first-order

A new theory that contradicted the old one everywhere would be a theory about a different universe. The correct relationship is subtler: relativity must *reduce* to Galileo where Galileo was tested. Expand and see.

By the binomial series (Chapter 0.3),

$$\gamma = (1 - \beta^2)^{-1/2} = 1 + \frac{1}{2}\beta^2 + \frac{3}{8}\beta^4 + \dots \quad (2.2.22)$$

so $\gamma = 1 + O(\beta^2)$. Substituting into (2.2.19) and keeping terms through first order in β :

$$x' = x - vt + O(\beta^2), \quad t' = t - \frac{vx}{c^2} + O(\beta^2). \quad (2.2.23)$$

Let $c \rightarrow \infty$ at fixed v, x, t — which is what "ordinary speeds" means — and the second term of t' vanishes too. What is left is $x' = x - vt, t' = t$: **the Galilean transformation**. Not approximately, not in spirit; exactly, as a limit.

That is the honest statement of the relationship, and it is worth saying what it means. Galilean relativity is not a mistake that got corrected. It is the leading term of an expansion, and the reason it looked like the whole truth for three centuries is that nobody could measure the next term. The same pattern will recur: Newtonian gravity is the leading term of general relativity (Chapter 3.6), classical mechanics is a limit of quantum mechanics (Chapter 4.6). Old theories almost never die; they get demoted to leading order.

But look closely at (2.2.23), because the two corrections do *not* enter at the same order, and that asymmetry explains the entire history of the subject:

- **Lengths and time rates** are wrong by a factor γ , i.e. by $\frac{1}{2}\beta^2$ — *second* order. At $\beta = 10^{-4}$ that is 5×10^{-9} , a part in two hundred million.
- **Simultaneity** is wrong by $-vx/c^2$ — *first* order in β , but multiplied by a distance. At $v = 30 \text{ km s}^{-1}$ and $x = 1 \text{ m}$ this is $3.3 \times 10^{-13} \text{ s}$. Small — but scale x up to a satellite orbit, $2 \times 10^7 \text{ m}$, and it becomes $0.9 \text{ } \mu\text{s}$, which is 260 m of light travel and would wreck a navigation system in an afternoon.

So the largest relativistic effect at low speed is the one about clocks at different *places*, and it is the one nobody notices, because noticing it requires two clocks far apart and a way to compare them. That is exactly the experiment nineteenth-century physics could not do and twenty-first-century engineering does continuously.

3 · Matrix form and the group property

Chapter 0.4 taught you to see a linear map as a matrix and composition as matrix multiplication. Do that here. Using $x^\mu = (ct, x, y, z)$ — time measured in metres, so all four coordinates carry the same units — (2.2.19) is

$$\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \Lambda \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}, \quad \Lambda^\mu{}_\nu = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (2.2.24)$$

The index placement $\Lambda^\mu{}_\nu$ — upper for the row, lower for the column — is not decoration; Chapter 2.4 explains exactly what promise it encodes, and derives this same matrix as a Jacobian $\partial x'^\mu / \partial x^\nu$. For now it is a matrix.

Two features are worth naming immediately. It is **symmetric**, which we will use in Problem 3. And its determinant is $\gamma^2 - \gamma^2\beta^2 = \gamma^2(1 - \beta^2) = 1$, so boosts preserve four-dimensional volume in spacetime — a fact that matters in statistical mechanics and in Chapter 5.6.

3.1 • Two boosts make a boost

Now the question that decides whether we have a theory or a formula. Boost from S to S' at β_1 ; then boost from S' to S'' at β_2 , along the same axis. Is the composite a boost? Multiply the matrices. Only the upper-left 2×2 block does anything:

$$\begin{aligned}\Lambda(\beta_2)\Lambda(\beta_1) &= \gamma_2 \begin{pmatrix} 1 & -\beta_2 \\ -\beta_2 & 1 \end{pmatrix} \gamma_1 \begin{pmatrix} 1 & -\beta_1 \\ -\beta_1 & 1 \end{pmatrix} \\ &= \gamma_1\gamma_2 \begin{pmatrix} 1 + \beta_1\beta_2 & -(\beta_1 + \beta_2) \\ -(\beta_1 + \beta_2) & 1 + \beta_1\beta_2 \end{pmatrix}.\end{aligned}\tag{2.2.25}$$

Pull out the common factor $1 + \beta_1\beta_2$ and define

$$\beta_3 \equiv \frac{\beta_1 + \beta_2}{1 + \beta_1\beta_2}.\tag{2.2.26}$$

Then the product is $\gamma_1\gamma_2(1 + \beta_1\beta_2)$ times the matrix $\begin{pmatrix} 1 & -\beta_3 \\ -\beta_3 & 1 \end{pmatrix}$, and it is a boost provided that prefactor equals $\gamma_3 = (1 - \beta_3^2)^{-1/2}$. It does — the grind box does the algebra — so

$$\Lambda(\beta_2)\Lambda(\beta_1) = \Lambda\left(\frac{\beta_1 + \beta_2}{1 + \beta_1\beta_2}\right).\tag{2.2.27}$$

▼ Grind box — the prefactor really is γ_3 , and one identity that keeps reappearing

Compute $1 - \beta_3^2$ from (2.2.26):

$$1 - \beta_3^2 = \frac{(1 + \beta_1\beta_2)^2 - (\beta_1 + \beta_2)^2}{(1 + \beta_1\beta_2)^2}.$$

Expand the numerator and watch the cross terms die:

$$\begin{aligned}(1 + ab)^2 - (a + b)^2 &= 1 + 2ab + a^2b^2 - a^2 - 2ab - b^2 \\ &= 1 - a^2 - b^2 + a^2b^2 = (1 - a^2)(1 - b^2).\end{aligned}$$

So

$$\sqrt{1 - \beta_3^2} = \frac{\sqrt{1 - \beta_1^2} \sqrt{1 - \beta_2^2}}{1 + \beta_1 \beta_2} = \frac{1}{\gamma_1 \gamma_2 (1 + \beta_1 \beta_2)},$$

hence $\gamma_3 = \gamma_1 \gamma_2 (1 + \beta_1 \beta_2)$, which is the prefactor. ✓

Keep that identity. $(1 + ab)^2 - (a + b)^2 = (1 - a^2)(1 - b^2)$ — and its sibling $(1 - ab)^2 - (a - b)^2 = (1 - a^2)(1 - b^2)$ — will do the work again in §5 (proving you cannot reach c by combining sub- c speeds) and once more in §6, where it will turn out to be the hyperbolic addition formula in disguise. Three appearances of one factorisation is usually a sign that something structural is going on, and it is.

The other group axioms. $\Lambda(0) = I$ is the identity. $\Lambda(-\beta)$ is the inverse, since (2.2.26) with $\beta_2 = -\beta_1$ gives $\beta_3 = 0$. Associativity is free — matrix multiplication is associative. So the boosts along a fixed axis form a **one-parameter group**, and §6 finds the parameter that makes it look like one.

This matters more than it looks. Closure is not automatic: it is a statement that the set of transformations is a self-consistent family, that "the frames of physics" is a coherent notion, and that no chain of boosts can ever take you outside the theory. Had the composition of two boosts failed to be a boost, the whole scheme would have been incoherent.

But look hard at (2.2.26). The composed parameter is **not** $\beta_1 + \beta_2$. Boost by $0.5c$ and then by $0.5c$ again and you get

$$\beta_3 = \frac{0.5 + 0.5}{1 + 0.25} = \frac{1}{1.25} = 0.8, \quad (2.2.28)$$

not 1. Two half-light-speed boosts give eight tenths, and no finite chain of sub- c boosts will ever reach c — which §5 proves properly and §6 explains. Velocity, it turns out, is simply the wrong variable to be adding.

⚠ ONE CAVEAT, FLAGGED NOW AND PAID IN PROBLEM 3

Everything above was for boosts along *one fixed axis*. Boosts along different axes do **not** compose into a pure boost: the product picks up a spatial rotation, the *Wigner rotation*. Problem 3 derives that the effect must exist, in two lines, from the fact that (2.2.24) is a symmetric matrix. The physical consequence — Thomas precession — is a measurable contribution to atomic fine structure, and it is a genuinely relativistic effect with no Newtonian ancestor at all.

4 • The three consequences

Now we cash it in. Three results, and every one of them is a two-line substitution into (2.2.19). The difficulty in this material is never the algebra; it is knowing *which two events* you are transforming. So each subsection begins by stating the operational question — who measures what, with which apparatus, and using which pair of events — before any symbol moves.

4.1 · Relativity of simultaneity

The operational question. Two firecrackers explode. An observer in S , using clocks that have been synchronised throughout S , records both explosions at the same reading. What do the clocks of S' — synchronised throughout S' — record?

Take two events with $\Delta t = 0$ and $\Delta x \neq 0$ in S . From (2.2.19), applied to the separation (legitimate because the transformation is linear, which is what §1 bought us):

$$c \Delta t' = \gamma(c \Delta t - \beta \Delta x) = -\gamma\beta \Delta x \quad \implies \quad \Delta t' = -\frac{\gamma\beta \Delta x}{c} \neq 0. \quad (2.2.29)$$

Events simultaneous in S are not simultaneous in S' , unless they happen at the same place. The sign matters and is worth reading off: if $\Delta x > 0$ — the second event is further along the direction of S' 's motion — then $\Delta t' < 0$, so in S' the leading event happened *first*. **The frame that is moving forward sees the forward event as earlier.**

A concrete case

A train of proper length 300 m passes a platform at $\beta = 0.6$, so $\gamma = 1.25$. Lightning strikes both ends of the train — simultaneously, as judged on the platform.

In the platform frame S the train is contracted (§4.3) to $300/1.25 = 240$ m, so the two strike events have $\Delta x = 240$ m, $\Delta t = 0$. On the train,

$$\Delta t' = -\frac{\gamma\beta\Delta x}{c} = -\frac{1.25 \times 0.6 \times 240}{2.998 \times 10^8} \text{ s} = -6.00 \times 10^{-7} \text{ s} = -600 \text{ ns}, \quad (2.2.30)$$

and $\Delta x' = \gamma\Delta x = 1.25 \times 240 = 300$ m ✓ — the train's own length, as it must be. So on the train, the front is struck 600 ns *before* the rear. Not "appears to be": is, in every sense that the train's own clocks can express.

Check it without the transformation, as a sanity test. Both flashes travel at c in the platform frame and meet at the midpoint of the two strike locations. Meanwhile the train's midpoint is moving toward the front strike, so it runs into the front flash first. Now argue in the train frame: the train observer sits midway between the two *scorch marks on the train*, both flashes travel to them at c over equal distances, and the front flash arrives first. Equal distances, equal speeds, different arrival times — therefore different emission times. The front strike happened first. Two independent routes, one answer. ✓

WHY THIS ONE GOES FIRST

Time dilation and length contraction are not independent phenomena sitting alongside the failure of simultaneity. They are *consequences* of it. Every apparent paradox in the rest of this chapter — and in the rest of the subject — is resolved by finding the place where somebody assumed two distant events had a frame-independent time order.

Readers who meet dilation and contraction first almost never recover, because they acquire a picture in which clocks and rulers are mysteriously defective while "now" remains universal. It does not. There is no universal "now". There is a family of them, one per state of motion, and they cut through spacetime at different angles. Hold that and nothing that follows can confuse you.

4.2 · Time dilation

The operational question. A single clock rides along with S' and ticks twice. The elapsed time it reads between the ticks is $\Delta\tau$ — this is called the clock's **proper time**, and it is what that particular clock actually displays. What is the elapsed time between those same two ticks according to S , where the reading must be taken off *two different* synchronised clocks, one at each tick's location?

The clock is at rest in S' , so the two ticks have $\Delta x' = 0$ and $\Delta t' = \Delta\tau$. We want Δt , so use the inverse transformation from the grind box in §2.3:

$$c \Delta t = \gamma (c \Delta t' + \beta \Delta x') = \gamma c \Delta\tau \quad \implies \quad \boxed{\Delta t = \gamma \Delta\tau} \quad (2.2.31)$$

Since $\gamma \geq 1$, more time elapses in S than the moving clock records. A moving clock runs slow, by exactly γ . Note the asymmetry in the *apparatus*, which is where all the content is: one clock on one side, two clocks on the other.

The symmetric version, and why it is not a contradiction

Run the identical argument the other way. A clock at rest in S ticks twice, $\Delta x = 0$, $\Delta t = \Delta\tau_S$. Then (2.2.19) gives directly $c\Delta t' = \gamma(c\Delta\tau_S - 0)$, so $\Delta t' = \gamma\Delta\tau_S$. **Each observer finds the other's clock slow, by the same factor.** Both statements are true. Both are derived from the same transformation. There is no contradiction, and here is the accounting that shows it.

Set $\beta = 0.6$, $\gamma = 1.25$. In S there are two clocks, C_1 at $x = 0$ and C_2 at $x = 12$ light-seconds, synchronised in S . A traveller carrying clock C' flies from C_1 to C_2 .

Told by S : the trip takes $\Delta t = 12/0.6 = 20$ s. The traveller's clock, moving, runs slow, and reads $20/1.25 = 16$ s on arrival. C_2 reads 20 s.

Told by S' : the traveller is at rest and the whole S apparatus streams past at $0.6c$. The distance from C_1 to C_2 is contracted to $12/1.25 = 9.6$ light-seconds, so it takes $9.6/0.6 = 16$ s — matching the traveller's own clock, obviously, since it is his own clock. During those 16 seconds the S clocks, being the moving ones now, tick only $16/1.25 = 12.8$ s.

Contradiction? C_2 must read 20 s when the traveller arrives — that is a local coincidence, two objects at the same place, and no one can disagree about it. Yet S' says the S clocks advanced only 12.8 s. The missing 7.2 s is exactly the

simultaneity offset. At the moment the traveller passes C_1 (call it $t' = 0$), the S clocks that S' regards as simultaneous satisfy $t = \beta x / c$, so C_2 — sitting at $x = 12$ light-seconds — already reads

$$t_{C_2} = \frac{\beta x}{c} = 0.6 \times 12 = 7.2 \text{ s.} \quad (2.2.32)$$

In the traveller's reckoning, C_2 was never synchronised with C_1 in the first place: it was 7.2 s ahead before the trip began. So the books balance:

$$\underbrace{7.2}_{\text{head start}} + \underbrace{12.8}_{\text{ticked during the trip}} = \underbrace{20.0}_{\text{what } C_2 \text{ reads}} \quad \checkmark \quad (2.2.33)$$

Both frames agree on every local coincidence — which clock reads what when it is next to which other clock. They disagree only about the synchronisation of distant clocks, which is not an observable but a convention imposed frame by frame. The symmetry of time dilation is *consistent precisely because* simultaneity fails, and the failure is exactly the right size. It could not have been any other size: (2.2.29) and (2.2.31) came out of the same matrix.

△ WHY THIS ISN'T OBVIOUS

"Each sees the other's clock run slow" sounds like $a < b$ and $b < a$. It is not, because the two statements are not comparisons of the same things. Each is a comparison of *one* clock against *a pair* of clocks synchronised in the other frame — and the two frames use different pairs, synchronised differently. Strip out the words and write down which events are being compared, and the appearance of contradiction evaporates every single time.

Here is the test to apply whenever you feel the vertigo: **ask whether the disputed claim is about a local coincidence.** "Clock C' reads 16 while sitting next to C_2 , which reads 20" is a coincidence — two objects, one place, one event — and every frame agrees on it. "Clock C' reads 16 while C_2 , twelve light-seconds away, reads 20" is not; it invokes a simultaneity convention, and frames are entitled to differ. All the paradoxes are manufactured by smuggling the second kind of statement in wearing the clothes of the first.

4.3 · Length contraction

The operational question, and it is the whole difficulty. A rod lies at rest in S' along the x' -axis. Its ends are at x'_L and x'_R , always, so its length in S' is unambiguous: $L_0 = x'_R - x'_L$, called the **proper length**. Now ask S for the rod's length. In S the rod is moving, so "where its ends are" is a function of time, and to get a length you must locate *both ends at the same moment*. At the same moment according to whom? To S — that is what it means for S to measure a length.

So a length measurement is a pair of events (locate left end, locate right end) constrained by $\Delta t = 0$ in the *measuring frame*. The moment you write that down, the answer is forced, because the constraint is a simultaneity constraint and §4.1 already told you those are frame-dependent.

Apply (2.2.19) to the pair. In S : $\Delta x = L$ (what we want) and $\Delta t = 0$ (the definition of measuring). In S' : $\Delta x' = L_0$, since the rod is at rest there and it does not matter when you look. Then

$$\Delta x' = \gamma(\Delta x - \beta c \Delta t) = \gamma \Delta x \quad \implies \quad L_0 = \gamma L, \quad (2.2.34)$$

$$\boxed{L = \frac{L_0}{\gamma} = L_0 \sqrt{1 - \beta^2} \leq L_0.} \quad (2.2.35)$$

A moving rod is short, by γ , along its direction of motion only ($y' = y, z' = z$ means nothing happens across it). And Chapter 2.1's loose end is tied: this is precisely the $\sqrt{1 - \beta^2}$ that FitzGerald and Lorentz had to *postulate* to explain Michelson–Morley. Here it is a theorem.

It follows from simultaneity — watch

The derivation used $\Delta t = 0$, which by (2.2.29) means the two measurement events are *not* simultaneous in S' :

$$\Delta t' = -\frac{\gamma\beta \Delta x}{c} = -\frac{\gamma\beta L}{c} = -\frac{\beta L_0}{c}. \quad (2.2.36)$$

From the rod's own frame, S located the right end first and the left end $\beta L_0/c$ later. Since the rod is not going anywhere in S' , that sloppiness costs S nothing — the ends are where they always are — which is why S still gets a definite answer. But it is exactly why the answer is not L_0 . To see that the choice of event pair is doing all the work, redo the calculation with a pair that is simultaneous in S' instead:

$$\begin{aligned} \Delta t' = 0 &\implies c \Delta t = \beta \Delta x \\ \implies \Delta x' &= \gamma(\Delta x - \beta^2 \Delta x) = \frac{\Delta x}{\gamma} \implies \Delta x = \gamma \Delta x'. \end{aligned} \quad (2.2.37)$$

Same rod, same frames, different pair of events, and the answer flips from L_0/γ to γL_0 . **The contraction is not a property of the rod. It is a property of the measurement**, and specifically of whose simultaneity slices the measurement uses. Nothing was squeezed. Two observers sliced the same four-dimensional object at different angles and got different cross-sections, exactly as two people slicing the same sausage at different angles get ellipses of different lengths. Chapter 2.3 makes that picture precise; here the algebra already contains it.

△ WHY THIS ISN'T OBVIOUS — CONTRACTION IS NOT "LOOKING SQUASHED"

Length contraction is a statement about a *simultaneous measurement*. It is not a statement about what a camera records, and the two differ, because light from different parts of an object takes different times to reach the lens. A photograph does not capture a simultaneity slice; it captures the events whose light arrives together.

Work out the difference for a cube of side L flying past at β , seen from far away, edge-on. Light from the rear face must set out earlier than light from the front face — earlier by L/c , the extra distance — in order to arrive at the camera together. During that L/c the cube moves a distance βL , so the rear face is *displaced sideways* in the image by βL ; you can see round the back of the cube.

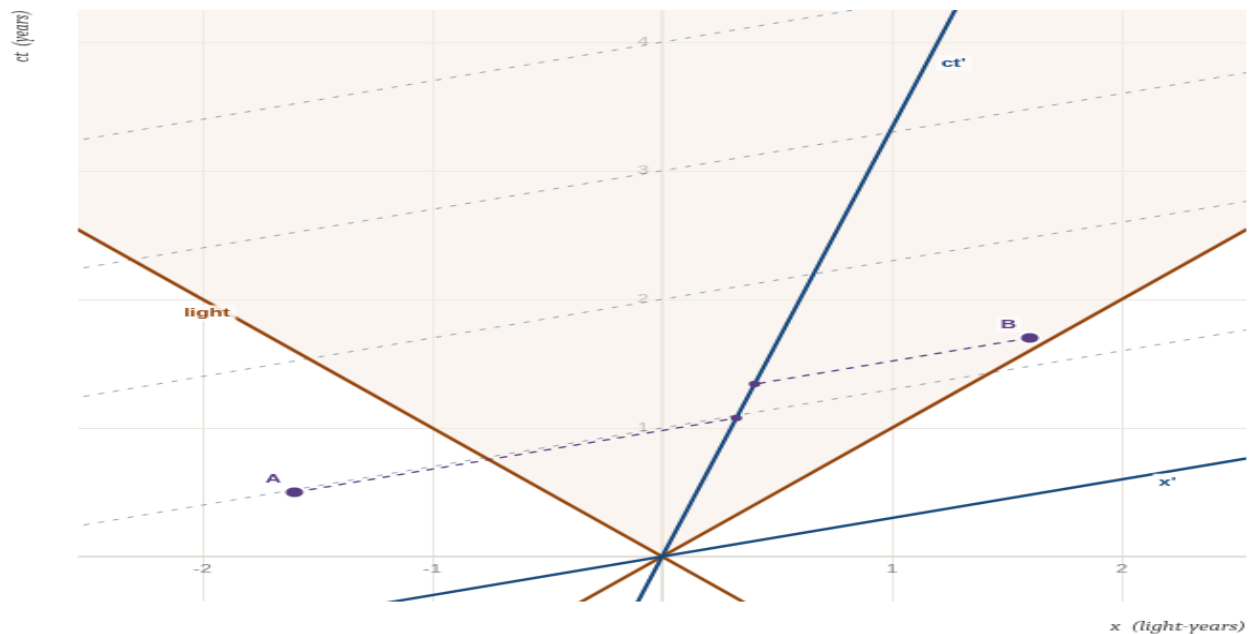
Meanwhile the front face, measured simultaneously, is contracted to $L\sqrt{1 - \beta^2}$. Now compare with a cube that is not moving at all but has been *rotated* by an angle θ : its front face projects to $L \cos \theta$ and one side face becomes visible with projected width $L \sin \theta$. Match them:

$$\sin \theta = \beta, \quad \cos \theta = \sqrt{1 - \beta^2} \quad \checkmark$$

The photograph of a rapidly moving cube is the photograph of a *rotated* cube, turned by $\theta = \arcsin \beta$.

■ The general theorem — that a small object of any shape appears rotated rather than contracted, and that a sphere always photographs as a perfectly circular disc — is the Terrell–Penrose result of 1959; the cube calculation above is the whole idea, and the general case is bookkeeping over solid angle.

So: contraction is real and it is not visual. If someone shows you a picture of a squashed spaceship, the picture is wrong. The spaceship *is* shorter, measured; it does not *look* shorter, photographed.



boost β

$\beta = 0.375$

$\beta = 0.8$

twin worldlines

$\beta = 0.300$ $\gamma = 1.04828$ $\phi = 0.30952$ (light lines fixed at 45°)

S : $\Delta t = +1.2000$ yr (A first) S' : $\Delta t' = +0.2516$ yr (A first)

twin worldlines: off

The whole chapter on one diagram. Coordinates of the unprimed frame S : x across, ct up. *Inside this figure only*, units are chosen with $c = 1$ — years for time, light-years for distance — so light travels one unit of distance per unit of time and the light lines sit at exactly 45° . Everything else in the book keeps c explicit. **(i)** The orange lines are the light cone through the origin. Drag β and watch them: they do not move. That is Postulate 2, drawn. **(ii)** The blue lines are the primed axes — ct' is the worldline of the S' origin ($x = \beta ct$) and x' is the set of events S' calls simultaneous with the origin ($ct = \beta x$). They scissor *symmetrically* toward the light line, which is the geometric content of (2.2.19): the same β appears in both equations. The faint dashed lines parallel to x' are S' 's other simultaneity slices — its "nows". **(iii)** Two spacelike-separated events, A and B , with A earlier in S by 1.2 yr. The dashed purple lines drop each event onto the ct' axis along S' 's simultaneity slices. Push β past 0.375 and the order reverses: $\Delta t'$ changes sign, and B — which happened later for S — happens earlier for S' . Nothing has been signalled; no cause has overtaken an effect. These two events cannot influence each other at all, which is exactly the licence that lets their order be frame-dependent. **(iv)** The twin toggle draws two worldlines from $(0, 0)$ to $(0, 4 \text{ yr})$: the straight one at $x = 0$, and the bent one that flies out at the current β for two years and returns. Proper time along each is computed numerically by summing $\sqrt{\Delta t^2 - \Delta x^2/c^2}$ over 8000 steps along the path — no formula is used — and the bent path always accumulates less. The green dashed lines are the traveller's simultaneity slices immediately before and immediately after the turn; where they cross $x = 0$ they mark the jump in the stay-at-home twin's clock that §7.1 is about.

5 · Velocity addition

An object moves at velocity u along x in frame S . What velocity u' does S' assign it? Galileo says $u' = u - v$ — the velocity-composition rule of Chapter 2.1 §1.2 — and it is exactly what Postulate 2 refuses to accept for light. So derive

the replacement.

Velocity is a ratio of displacements, so take differentials of (2.2.19) — legitimate because the coefficients are constants, so the differentials obey the same linear relations as the coordinates:

$$dx' = \gamma(dx - v dt), \quad dt' = \gamma\left(dt - \frac{v dx}{c^2}\right). \quad (2.2.38)$$

Divide. The factors of γ cancel — a small mercy, and a hint that the result depends on β less violently than one might fear:

$$u' = \frac{dx'}{dt'} = \frac{dx - v dt}{dt - v dx/c^2} = \frac{\frac{dx}{dt} - v}{1 - \frac{v}{c^2} \frac{dx}{dt}}, \quad (2.2.39)$$

$$\boxed{u' = \frac{u - v}{1 - \frac{uv}{c^2}}} \quad \text{equivalently} \quad \beta_{u'} = \frac{\beta_u - \beta_v}{1 - \beta_u \beta_v}. \quad (2.2.40)$$

The dimensionless form is (2.2.26) with a sign flipped, as it should be: composing two boosts and transforming a velocity are the same operation seen from two sides.

5.1 · The postulate reproduces itself

Set $u = c$ and turn the crank:

$$u' = \frac{c - v}{1 - \frac{cv}{c^2}} = \frac{c - v}{1 - \frac{v}{c}} = \frac{c(c - v)}{c - v} = c. \quad (2.2.41)$$

For *any* v with $|v| < c$. This is a real check, not a tautology dressed up. We imposed the invariance of c only for a pulse emitted from the origin at $t = 0$ along $\pm x$; the transformation that resulted now certifies that *any* object moving at c in S moves at c in every frame — anywhere, any time, either direction. A theory that failed this would have been inconsistent with its own founding assumption.

5.2 · You cannot get to c by adding speeds

Claim. If $|\beta_u| < 1$ and $|\beta_v| < 1$ then $|\beta_{u'}| < 1$.

Proof. First, the denominator never vanishes: $|\beta_u \beta_v| < 1$, so $1 - \beta_u \beta_v > 0$. Now compute, using the identity from the §3 grind box in its second form, $(1 - ab)^2 - (a - b)^2 = (1 - a^2)(1 - b^2)$:

$$1 - \beta_{u'}^2 = \frac{(1 - \beta_u \beta_v)^2 - (\beta_u - \beta_v)^2}{(1 - \beta_u \beta_v)^2} = \frac{(1 - \beta_u^2)(1 - \beta_v^2)}{(1 - \beta_u \beta_v)^2}. \quad (2.2.42)$$

Every factor on the right is strictly positive, so $1 - \beta_{u'}^2 > 0$, so $|\beta_{u'}| < 1$. ■

Read (2.2.42) once more, because it says more than the claim. It says the "distance from c ", measured by $1 - \beta^2$, is *multiplicative* under composition: combine two sub-light speeds and the product of their deficits is what survives. You can multiply small positive numbers together forever and never reach zero. The light barrier is not a wall someone installed; it is the statement that a product of positives is positive.

5.3 · And at ordinary speeds, Galileo

For $uv/c^2 \ll 1$, expand the denominator (Chapter 0.3):

$$u' = (u - v) \left(1 + \frac{uv}{c^2} + \dots \right) = u - v + \frac{uv(u - v)}{c^2} + \dots \quad (2.2.43)$$

Two cars approaching each other at 30 m s^{-1} each ($u = +30, v = -30$): the Galilean answer is 60 m s^{-1} and the correction is

$$\frac{uv(u - v)}{c^2} = \frac{(30)(-30)(60)}{8.99 \times 10^{16}} = -6.0 \times 10^{-13} \text{ m s}^{-1}, \quad (2.2.44)$$

a relative error of one part in 10^{14} . This is why the Galilean rule survived so long unchallenged: on a motorway it is wrong in the fourteenth decimal place.

△ WHY THIS ISN'T OBVIOUS — SOME SPEEDS MAY EXCEED c , AND IT IS FINE

Nothing above forbids *coordinate* speeds greater than c . What is forbidden is the transport of matter, energy or information faster than c . Two examples, both computed.

A spotlight on a distant wall. Rotate a laser at angular rate ω ; the spot on a wall at distance R moves at $v_{\text{spot}} = \omega R$, with no upper bound. Point a laser at the Moon ($R = 3.84 \times 10^8$ m) and spin it at one revolution per second, $\omega = 2\pi \text{ s}^{-1}$:

$$v_{\text{spot}} = 2\pi \times 3.84 \times 10^8 = 2.41 \times 10^9 \text{ m s}^{-1} = 8.05 c.$$

No rule is broken, because the spot is not a thing. The photons landing at one place and the photons landing at the next place travelled independently from the laser; nothing went sideways. You cannot send a message along the wall this way — each point of the wall learns only about the laser, never about its neighbours.

A pair of scissors. Two straight edges crossing at a small angle θ , one sliding toward the other at speed v . The moving edge is $y = x \tan \theta + vt$, so the intersection with $y = 0$ sits at $x = -vt / \tan \theta$ and travels at $v / \tan \theta$ — unbounded as $\theta \rightarrow 0$. With $v = 100 \text{ m s}^{-1}$ and $\theta = 10^{-7}$ rad the crossing point moves at $3.3 c$. Again nothing is transported; the intersection is a geometric coincidence, not an object, and a real pair of scissors would in any case not be rigid — closing the handles sends a wave down the blade at the speed of sound in steel.

The rule to carry: **ask what is being transported**. If the answer is "nothing", any speed is allowed.

6 · Rapidity — the parameter that behaves

Velocities do not add. Something must, because §3 showed that boosts along an axis form a one-parameter group, and a one-parameter group has an additive parameter almost by definition: composing two elements should add their labels, the way rotating by 30° and then 40° gives 70° . We simply chose a bad label. Find the good one.

The composition rule (2.2.26),

$$\beta_3 = \frac{\beta_1 + \beta_2}{1 + \beta_1 \beta_2}, \quad (2.2.45)$$

is not a formula anyone would invent — but it is a formula everyone has seen. It is the addition theorem for the hyperbolic tangent. So define the **rapidity** ϕ by

$$\beta = \tanh \phi, \quad \text{equivalently} \quad \phi = \text{artanh } \beta = \frac{1}{2} \ln \frac{1 + \beta}{1 - \beta}. \quad (2.2.46)$$

This is a legitimate change of variable: $\tanh$ maps $\mathbb{R}$ one-to-one onto $(-1, 1)$, exactly the allowed range of β , and it is smooth with a smooth inverse. Every subluminal velocity has exactly one rapidity, and every real number is somebody's rapidity.

6.1 • What γ becomes

Divide the identity $\cosh^2 \phi - \sinh^2 \phi = 1$ by $\cosh^2 \phi$ to get $1 - \tanh^2 \phi = 1 / \cosh^2 \phi$. Hence

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}} = \frac{1}{\sqrt{1 - \tanh^2 \phi}} = \cosh \phi, \quad \gamma\beta = \cosh \phi \tanh \phi = \sinh \phi. \quad (2.2.47)$$

Substituting into (2.2.24), the boost is

$$\begin{pmatrix} ct' \\ x' \end{pmatrix} = \begin{pmatrix} \cosh \phi & -\sinh \phi \\ -\sinh \phi & \cosh \phi \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}, \quad (2.2.48)$$

and the transformation reads $ct' = ct \cosh \phi - x \sinh \phi$, $x' = x \cosh \phi - ct \sinh \phi$.

6.2 • Rapidities add

Multiply two of them:

$$\begin{aligned} & \begin{pmatrix} \cosh \phi_2 & -\sinh \phi_2 \\ -\sinh \phi_2 & \cosh \phi_2 \end{pmatrix} \begin{pmatrix} \cosh \phi_1 & -\sinh \phi_1 \\ -\sinh \phi_1 & \cosh \phi_1 \end{pmatrix} \\ &= \begin{pmatrix} c_1 c_2 + s_1 s_2 & -(s_1 c_2 + c_1 s_2) \\ -(c_1 s_2 + s_1 c_2) & s_1 s_2 + c_1 c_2 \end{pmatrix}, \end{aligned} \quad (2.2.49)$$

writing $c_i = \cosh \phi_i$, $s_i = \sinh \phi_i$. The hyperbolic addition formulas — $\cosh(a + b) = \cosh a \cosh b + \sinh a \sinh b$ and $\sinh(a + b) = \sinh a \cosh b + \cosh a \sinh b$ — turn that into

$$\boxed{\Lambda(\phi_2) \Lambda(\phi_1) = \Lambda(\phi_1 + \phi_2).} \quad (2.2.50)$$

Rapidities add. Plainly, exactly, with no correction term. And taking $\tanh$ of both sides recovers (2.2.26), since $\tanh(\phi_1 + \phi_2) = (\tanh \phi_1 + \tanh \phi_2) / (1 + \tanh \phi_1 \tanh \phi_2)$ — so §5 and §6 are the same statement written in two variables.

Check the arithmetic of (2.2.28) in the new currency: $\operatorname{artanh}(0.5) = 0.549306$, twice that is $1.098612 = \ln 3$, and $\tanh(\ln 3) = (3 - \frac{1}{3}) / (3 + \frac{1}{3}) = 0.8 \checkmark$. Two boosts of $0.5c$ do not give c ; they give rapidity 2×0.5493 , which is velocity $0.8c$.

THE MORAL

Velocities "fail to add" for the same reason angles-as-slopes fail to add. Compose two rotations and their *angles* add; their *slopes* combine by $(m_1 + m_2)/(1 - m_1 m_2)$, which nobody finds shocking because nobody expected slopes to be the natural parameter. Velocity is the slope of a worldline. Rapidity is its angle. We spent three hundred years adding the slopes.

And the change of variable makes the light barrier trivial rather than mysterious. Rapidity runs over the whole of $\mathbb{R}$ and adds without limit; you can boost forever. But $\beta = \tanh \phi$ saturates: $\phi = 1$ gives $\beta = 0.762$, $\phi = 3$ gives 0.995, $\phi = 10$ gives 0.9999999959. The barrier is not a barrier at all — it is the horizontal asymptote of $\tanh$, seen from inside a badly chosen coordinate.

TWO FORWARD POINTERS, FLAGGED ▀

(a) This is a Lie group in embryo. The boosts along one axis are a family $\Lambda(\phi)$ with $\Lambda(0) = I$, $\Lambda(\phi_2)\Lambda(\phi_1) = \Lambda(\phi_1 + \phi_2)$ and Λ smooth in ϕ — a smooth one-parameter group. ▀ Chapter 6.1 shows that any such family is $\Lambda(\phi) = \exp(\phi K)$ for a fixed matrix K , its *generator*, obtained by differentiating at the identity. Here $K = \frac{d\Lambda}{d\phi}|_0 = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$, and indeed $K^2 = -I$, so $\exp(\phi K) = I \cosh \phi + K \sinh \phi$, which is (2.2.48) exactly. The full Lorentz group has six such parameters — three boosts, three rotations — and is called $SO(1, 3)$.

(b) The hyperbolic functions are the metric's minus sign, surfacing. A rotation in the xy plane is $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ and preserves $x^2 + y^2$. A boost in the (ct, x) plane is (2.2.48) and preserves $c^2 t^2 - x^2$, per (2.2.20). Circular functions preserve a sum of squares; hyperbolic functions preserve a difference. The two matrices differ in exactly one respect: the rotation's off-diagonal entries carry opposite signs, the boost's carry the same sign — the boost matrix is symmetric, the rotation's off-diagonal part is not. That single sign is the $-$ in $\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$. Chapter 2.3 makes this geometric: a boost is a rotation through an imaginary angle, or better, an honest rotation in a geometry whose circles are hyperbolas.

▼ Grind box — hyperbolic functions from scratch, in case they are rusty

Definitions, and everything else follows:

$$\cosh \phi = \frac{e^\phi + e^{-\phi}}{2}, \quad \sinh \phi = \frac{e^\phi - e^{-\phi}}{2}, \quad \tanh \phi = \frac{\sinh \phi}{\cosh \phi}.$$

The Pythagorean identity. Multiply out:

$$\cosh^2 \phi - \sinh^2 \phi = \frac{(e^\phi + e^{-\phi})^2 - (e^\phi - e^{-\phi})^2}{4} = \frac{4e^\phi e^{-\phi}}{4} = 1.$$

Compare $\cos^2 + \sin^2 = 1$: the point $(\cos \theta, \sin \theta)$ traces a circle, the point $(\cosh \phi, \sinh \phi)$ traces a hyperbola. That hyperbola is Chapter 2.3's central object.

Addition. Multiply exponentials:

$$\begin{aligned}\cosh a \cosh b + \sinh a \sinh b &= \frac{1}{4} \left[(e^a + e^{-a})(e^b + e^{-b}) + (e^a - e^{-a})(e^b - e^{-b}) \right] \\ &= \frac{1}{4} \left[2e^{a+b} + 2e^{-a-b} \right] = \cosh(a+b).\end{aligned}$$

The sinh formula goes the same way. Dividing them gives the tanh addition theorem, which is (2.2.45).

The inverse. Solve $\beta = \tanh \phi$ for ϕ . Writing $w = e^{2\phi}$,

$$\beta = \frac{e^\phi - e^{-\phi}}{e^\phi + e^{-\phi}} = \frac{w - 1}{w + 1} \implies w = \frac{1 + \beta}{1 - \beta} \implies \phi = \frac{1}{2} \ln \frac{1 + \beta}{1 - \beta}.$$

Useful values: $\operatorname{artanh}(0.5) = 0.5493$, $\operatorname{artanh}(0.8) = 1.0986 = \ln 3$, $\operatorname{artanh}(0.9) = 1.4722$, $\operatorname{artanh}(0.99) = 2.6467$. Note how slowly rapidity grows once β is close to 1 — going from $0.9c$ to $0.99c$ costs about 1.17 of rapidity, roughly the same as going from rest to $0.8c$. In Chapter 2.5 that observation acquires teeth: the energy needed is what grows without bound.

Derivatives. $\cosh' = \sinh$, $\sinh' = \cosh$, $\tanh' = 1/\cosh^2 = 1 - \tanh^2$, all immediate from the exponential definitions. Problem 4 uses the last one.

7 · The paradoxes, dismantled

There are perhaps a dozen famous special-relativity paradoxes and they are all the same paradox. Here are the two that matter, worked to the bottom, and then the general statement.

7.1 · The twin paradox

The setup. Two twins, both aged 20. One stays on Earth. The other flies to a star 4 light-years away at $\beta = 0.8$ ($\gamma = 5/3$), turns round, and comes back. Earth-frame arithmetic: the outbound leg takes $4/0.8 = 5$ years, the trip takes 10 years, and by (2.2.31) the traveller's clock — moving — records $10/\gamma = 10 \times 0.6 = 6$ years. The traveller returns aged 26, the stay-at-home is 30.

The apparent paradox, stated as sharply as it deserves. Motion is relative. From the traveller's point of view, *Earth* receded at $0.8c$ and came back. By §4.2, each observer finds the other's clock slow — symmetrically, by the same γ . So the traveller should equally conclude that Earth's clock ran slow throughout: his own 6 years, divided by $\gamma = 5/3$, gives 3.6 years for the Earth twin, who should therefore be the *younger* one on his return. Both cannot be true: when the two stand next to each other at the end, comparing wrinkles is a local coincidence, and there is one fact of the matter.

Where the symmetry fails. It fails at the turnaround, and it fails physically, not verbally. The stay-at-home twin occupies a single inertial frame for the whole story. The traveller does not: he is at rest in one inertial frame going out and a *different* one coming back. The switch is detectable from inside the ship without looking out of the window — an accelerometer registers it, coffee spills, the traveller is pressed into his seat. There is no symmetry to appeal to, because the two histories are physically different, and no amount of "motion is relative" makes them the same. One worldline is straight; the other has a kink.

That is the resolution in one sentence, and people rightly find it unsatisfying, because it explains why the twins *may* differ without showing where the missing 6.4 years went — the gap between the traveller's naive 3.6 and the actual 10. So track them.

Following the traveller's own accounting, year by year

Work in Earth-frame coordinates (ct, x) measured in light-years and years, with the departure at the origin. The turnaround is the event $T = (ct, x) = (5, 4)$.

Outbound leg. The traveller is at rest in the frame S' moving at $\beta = +0.8$. His own elapsed time is, from (2.2.19) applied to the event T ,

$$ct'_T = \gamma(ct_T - \beta x_T) = \frac{5}{3}(5 - 0.8 \times 4) = \frac{5}{3} \times 1.8 = 3 \text{ years.} \quad (2.2.51)$$

Which of Earth's events does he call simultaneous with his turnaround? Those with the same t' , i.e. $ct - \beta x = 1.8$. On Earth $x = 0$, so

$$ct_{\text{Earth}} = 1.8 \text{ years.} \quad (2.2.52)$$

Consistent with time dilation as the traveller sees it: his 3 years, divided by $\gamma = 5/3$, gives 1.8 years on the Earth clock. From out there, Earth is the moving one and Earth is running slow. ✓

Inbound leg. After the turn he is at rest in S'' moving at $\beta = -0.8$. Its simultaneity slices are $ct + \beta x = \text{const}$, and through the same turnaround event $T = (5, 4)$,

$$ct + 0.8x = 5 + 3.2 = 8.2 \quad \implies \quad ct_{\text{Earth}} = 8.2 \text{ years.} \quad (2.2.53)$$

The jump. An instant before the turn, the traveller's "now on Earth" is year 1.8. An instant after, it is year 8.2. Nothing happened to Earth; the traveller changed which slice of spacetime he calls "now", and the slice swung through Earth's worldline:

$$\Delta t_{\text{jump}} = 8.2 - 1.8 = 6.4 \text{ years} = \frac{2\beta L}{c} = 2 \times 0.8 \times 4. \quad (2.2.54)$$

Now the books balance exactly:

$$\underbrace{1.8}_{\text{outbound}} + \underbrace{6.4}_{\text{turnaround jump}} + \underbrace{1.8}_{\text{inbound}} = 10.0 \text{ years} \quad \checkmark \quad (2.2.55)$$

while the traveller's own clock reads $3 + 3 = 6$ years. Every year is accounted for. The traveller's claim "Earth's clock ran slow" was correct on *each leg*; what he cannot do is stitch the two legs' notions of "now" together and pretend they were one. The naive symmetric argument gave Earth $1.8 + 1.8 = 3.6$ years, and the 6.4 it was missing is exactly the span his simultaneity slice swept across Earth's worldline during the turn — the one moment at which he was not an inertial observer, and therefore the one moment at which his slices were not entitled to line up.

THE VERSION TO ACTUALLY REMEMBER

Forget clocks for a moment. Proper time is the *length of a worldline*: between two events, $\Delta\tau = \sqrt{\Delta t^2 - \Delta x^2/c^2}$ along each straight piece, and you add the pieces. Two different paths between the same two endpoints have different lengths. Nobody is scandalised that a detour through the next town adds mileage.

Check the twin numbers this way. Straight path: $\Delta\tau = \sqrt{10^2 - 0} = 10$ years. Bent path, two legs: $\sqrt{5^2 - 4^2} = 3$ each, total 6. Same endpoints, different lengths, no paradox — and no need to mention acceleration at all, except to say that a kink is what makes a path longer or shorter than a straight one.

The one genuinely strange thing is the sign. Because of the minus in $\Delta t^2 - \Delta x^2/c^2$, **the straight path is the one with the most proper time**, not the least. Detours in spacetime save you time rather than costing it. That inverted extremum is not a curiosity; it is the principle from which Chapter 3.1 derives the motion of freely falling bodies, and it is why a satellite's orbit is the path that maximises the proper time of the clock aboard it.

Turn on the twin toggle in the figure above and watch this happen. The traveller's two simultaneity slices are the green dashed lines through the turnaround; where they cross $x = 0$ they bracket exactly the jump. Drag β and the numbers move, but the three pieces always sum to the stay-at-home twin's total.

7.2 · The pole and the barn

The setup. A runner carries a pole of proper length 15 m through a barn of proper length 10 m with a door at each end, at $\beta = 0.8$ so $\gamma = 5/3$. In the barn's frame the pole is contracted to $15 \times 0.6 = 9$ m: it fits, with a metre to spare. The farmer shuts both doors simultaneously at the moment it is inside, and — this is the claim to be tested — the pole is briefly wholly enclosed.

The apparent paradox. In the runner's frame the pole is 15 m and the barn is contracted to $10 \times 0.6 = 6$ m. The pole is two and a half times the barn's length. It cannot possibly be enclosed. But "the pole was inside the barn with both doors shut" ought to be a fact, not an opinion.

The resolution. It is not a fact, because it is not a local coincidence. It is a statement about two events at different places — *front door shuts* and *back door shuts* — being simultaneous. By §4.1 that is frame-dependent, and here the disagreement is enormous: with $\Delta x = 10$ m,

$$c \Delta t' = -\gamma\beta \Delta x = -\frac{5}{3} \times 0.8 \times 10 = -\frac{40}{3} = -13.33 \text{ m}, \quad (2.2.56)$$

The two door-closings are 13.33 m of light travel — that is, 44.5 ns — apart in the runner's frame, with the far door going first. They do not shut together at all. The far door shuts and reopens long before the pole's nose gets there; much later, the near door shuts behind the pole's tail. The pole passes through a barn that is opening and closing around it like a slow shutter, and it is never enclosed for an instant.

Both frames are right, and — the decisive point — they agree on everything that is not a convention. Was the pole ever hit by a door? No, in both frames. Did the nose emerge from the far door before the far door shut? Problem 2 tabulates every event in both frames and the answer is the same in both. What they disagree on is only "did the two doors shut at the same time", which is not an observable but a labelling.

7.3 · The general lesson

EVERY SPECIAL-RELATIVITY PARADOX IS THE SAME PARADOX

Somewhere in the statement, someone has assumed simultaneity is absolute. Find that assumption and the paradox dissolves. There is no second mechanism.

The reliable procedure, which will get you through any of them:

1. **List the events.** Not objects, not "the rod" or "the ship" — *events*, each with a definite (t, x) in one frame. Anything that is not an event is not yet a physics question.
2. **Ask which claims are local coincidences.** Two things at the same place at the same time: a door hitting a pole, two clocks side by side, a twin shaking a hand. These are frame-independent and every frame must agree on them, full stop.
3. **Transform.** Apply (2.2.19) to each event and tabulate.
4. **Locate the smuggled assumption.** It will be a claim of the form "and at that moment, over there, ..." — a statement about distant simultaneity dressed as a fact.

Apply that to the twins (the smuggled claim: "while the traveller aged 3 years, Earth aged 1.8, and that was still true after the turn"), to the pole (the smuggled claim: "both doors shut at the same time" as an absolute), or to any of the others — the ladder and the trapdoor, the rigid rod pushed at one end, the submarine that both sinks and floats. Same disease, same cure, every time.

8 · Worked examples

WORKED EXAMPLE 1 — COSMIC-RAY MUONS, DONE IN BOTH FRAMES

Muons are produced by cosmic-ray collisions about 15 km up in the atmosphere. ▀ Their mean lifetime at rest is $\tau_0 = 2.2 \mu\text{s}$, measured in the laboratory, and they decay exponentially: the fraction surviving after a proper time τ is $e^{-\tau/\tau_0}$ (Chapter 0.1 §5 — same first-order kinetics as drug clearance). Take $\beta = 0.998$. What fraction reaches sea level, according to (a) Newtonian expectations and (b) relativity? Then redo (b) entirely in the muon's frame, where the atmosphere is contracted rather than the clock dilated, and check the two accounts agree.

Step 0 — the Lorentz factor.

$$\gamma = \frac{1}{\sqrt{1 - 0.998^2}} = \frac{1}{\sqrt{3.996 \times 10^{-3}}} = 15.82.$$

Step 1 — Newtonian. No dilation, no contraction: the muon crosses 15 km at $0.998c$, and its internal clock is the same as ours.

$$t = \frac{15\,000}{0.998 \times 2.998 \times 10^8} = 5.013 \times 10^{-5} \text{ s} = 50.13 \mu\text{s},$$

which is $50.13/2.2 = 22.79$ lifetimes. Surviving fraction:

$$e^{-22.79} = 1.27 \times 10^{-10}.$$

About one muon in eight billion. The predicted flux at sea level is, to any instrument ever built, zero. ▀ What is actually observed is a copious flux — of order one muon per square centimetre per minute at sea level. The Newtonian prediction is not slightly wrong.

Step 2 — relativity, told from the ground. The muon's internal clock is the moving one, so by (2.2.31) the proper time it experiences during our $50.13 \mu\text{s}$ is

$$\Delta\tau = \frac{\Delta t}{\gamma} = \frac{50.13}{15.82} = 3.169 \mu\text{s} = 1.441 \text{ lifetimes},$$

$$e^{-1.441} = 0.2368.$$

Nearly a quarter of them make it. The two predictions differ by a factor of 1.9×10^9 — which is why this is not a subtle test but a demonstration you can do with a scintillator on a bench, and why ▀ Rossi and Hall's 1941 mountain-versus-sea-level muon counts were decisive.

Step 3 — the same physics, told from the muon. This is the half that gets skipped, and it is the half that matters, because in the muon's own frame *nothing dilates its clock*. Its clock is at rest. It lives $2.2 \mu\text{s}$ on average, exactly as advertised — and it had better still reach the ground, because whether a given muon arrives is a local coincidence and frames are not allowed to disagree about it.

What is different in the muon's frame is the *atmosphere*, which is now the moving object. By (2.2.35), the 15 km column of air — proper length, since it is at rest relative to the ground — is contracted to

$$L = \frac{L_0}{\gamma} = \frac{15\,000}{15.82} = 948.2 \text{ m.}$$

That column rushes past the muon at $0.998c$, taking

$$\Delta t_{\text{muon}} = \frac{948.2}{0.998 \times 2.998 \times 10^8} = 3.169 \mu\text{s},$$

and the surviving fraction is $e^{-3.169/2.2} = e^{-1.441} = 0.2368$. **Identical.** ✓

What this teaches. The two frames disagree about *why* the muon survives — one says the clock ran slow, the other says the trip was short — and agree exactly on the observable, which is the number of muons per second landing on the detector. That is the correct relationship between frames in every problem in this chapter: mechanisms are frame-dependent, observables are not. Note also that neither account works without the other's ingredient: use dilation *and* contraction together and you would double-count by a factor of γ ; use neither and you get 10^{-10} . There is exactly one factor of γ in the problem, and each frame places it somewhere different.

WORKED EXAMPLE 2 — CLOSING SPEED VERSUS RELATIVE SPEED

In a laboratory, two particles fly directly toward each other, each at $0.9c$ as measured in the lab. (a) At what rate does the gap between them close, in lab coordinates? (b) What is the speed of one as measured in the rest frame of the other? (c) Reconcile.

(a) The closing rate. In lab coordinates, particle 1 is at $x_1 = -d/2 + 0.9ct$ and particle 2 at $x_2 = d/2 - 0.9ct$. The gap is $x_2 - x_1 = d - 1.8ct$, so

$$-\frac{d}{dt}(x_2 - x_1) = 1.8c.$$

The gap shrinks at $1.8c$. This is *correct* and no rule is violated. It is not the speed of anything; it is the rate of change of a difference of two coordinates, both measured in one frame. Nothing is transported at $1.8c$ — as in §5's spotlight, ask what is being carried, and the answer is nothing.

(b) The relative speed. This is a different question: what does particle 2's *own* frame measure for particle 1? Take S' to be particle 2's rest frame, so $\beta_v = -0.9$ (it moves in the $-x$ direction in the lab), and the object of interest has $\beta_u = +0.9$. Then by (2.2.40),

$$\beta_{u'} = \frac{\beta_u - \beta_v}{1 - \beta_u \beta_v} = \frac{0.9 - (-0.9)}{1 - (0.9)(-0.9)} = \frac{1.8}{1.81} = 0.994475.$$

So $u' = 0.9945c$. Not $1.8c$; not even c . Each particle sees the other approaching at 99.45% of light speed.

Cross-check with rapidity. $\text{artanh}(0.9) = 1.472219$, and the relative rapidity is the sum, $2 \times 1.472219 = 2.944439$. Then $\tanh(2.944439) = 0.994475$ ✓ — the same number, obtained by addition rather than by a formula with a denominator in it. This is §6 earning its keep.

(c) Reconciling. The two answers are answers to two different questions, and the language of everyday physics does not distinguish them because at low speed they coincide: for $\beta \ll 1$, $\beta_{u'} \approx \beta_u - \beta_v$, which is the closing rate. At high speed they separate dramatically. The rule: **a closing rate is a statement about one frame's coordinates and can be up to $2c$; a relative velocity is a statement about one object's rest frame and is always less than c .** Only the second one is a velocity in the sense that (2.2.40) and Chapter 2.5's dynamics care about.

Why the difference has teeth. In a collider the quantity that determines what you can produce is the energy in the centre-of-mass frame, which depends on the relative velocity, not the closing rate. Two beams at $0.9c$ head-on do not collide at $1.8c$ in any useful sense; they collide at a relative γ of $\cosh(2.944) = 9.53$, which is $\gamma_{\text{rel}} = \gamma_1 \gamma_2 (1 + \beta_1 \beta_2) = 2.294^2 \times 1.81 = 9.53$ ✓ — the composition law of §3, doing an accelerator physicist's arithmetic.

9 • Your turn

PROBLEM 1 — THE TRANSVERSE VELOCITY, WHICH CHANGES EVEN THOUGH $Y' = Y$

An object has velocity components (u_x, u_y) in S . Derive u'_x and u'_y in S' . Explain why $u'_y \neq u_y$ even though the y coordinate itself is untouched. Then check the formula on light: a pulse moving in the $+y$ direction at speed c in S — verify that S' also measures speed c , and find the angle by which the pulse's direction is tilted.

– Solution

Derivation. Take differentials of (2.2.19) including the transverse equation $dy' = dy$:

$$dx' = \gamma(dx - vdt), \quad dy' = dy, \quad dt' = \gamma\left(dt - \frac{v dx}{c^2}\right).$$

Divide dy' by dt' and then divide numerator and denominator by dt :

$$u'_y = \frac{dy}{\gamma(dt - vdx/c^2)} = \frac{u_y}{\gamma\left(1 - \frac{u_x v}{c^2}\right)}, \quad u'_x = \frac{u_x - v}{1 - \frac{u_x v}{c^2}}.$$

Why it changes. A velocity is a ratio, $\Delta y / \Delta t$. The numerator is indeed untouched — but the *denominator* is not. Time intervals transform, so a displacement that is unchanged, divided by a duration that is not, gives a changed rate. The lesson generalises far beyond this problem: an invariant numerator does not make an invariant quotient, which is exactly why Chapter 2.5 will insist on differentiating with respect to *proper* time rather than coordinate time when it builds the four-velocity.

The special case $u_x = 0$. Then $u'_x = -v$ and $u'_y = u_y / \gamma$: the transverse velocity is reduced by exactly γ , since the transverse displacement is unchanged while the time to make it is dilated.

Check on light. Put $u_x = 0$, $u_y = c$. Then $u'_x = -v$ and $u'_y = c / \gamma$, so

$$|u'|^2 = v^2 + \frac{c^2}{\gamma^2} = v^2 + c^2(1 - \beta^2) = v^2 + c^2 - v^2 = c^2.$$

Speed c ✓ — and note how the two pieces conspire: the v^2 gained along x is precisely the $c^2\beta^2$ lost along y . Postulate 2 again, checking itself in a direction we never imposed it.

The tilt. The direction makes an angle θ' with the y' -axis given by

$$\tan \theta' = \frac{|u'_x|}{u'_y} = \frac{v}{c/\gamma} = \gamma\beta, \quad \text{equivalently} \quad \sin \theta' = \beta.$$

This is **stellar aberration**: the apparent position of a star shifts by $\theta' \approx \beta$ as the Earth's velocity changes around its orbit, giving an annual ellipse of angular radius $\beta = 9.94 \times 10^{-5}$ rad, which in

arcseconds ($\times 206265$) is $20.5''$. Bradley measured it in 1727, and it was one of the constraints Chapter 2.1 §6.1 used to kill the fully dragged ether. Chapter 2.1's Problem 3 obtained $\theta \approx v/c$ from a first-order argument; $\sin \theta' = \beta$ is the exact statement, and the two agree to one part in 10^8 at the Earth's orbital speed. Note also that $\sin \theta' = \beta$ is the same relation as the Terrell–Penrose rotation angle in §4.3 — not a coincidence, since both are the statement that a boost tilts light rays.

PROBLEM 2 — THE POLE AND THE BARN, WITH EVERY EVENT TABULATED

Pole of proper length 15 m, barn of proper length 10 m with doors at $x = 0$ (entrance) and $x = 10$ m (exit), runner at $\beta = 0.8$ in the $+x$ direction. Use coordinates (ct, x) with both in metres. Let event A — the pole's tail reaching the entrance — be the origin of both frames. (a) Find the barn-frame times at which the pole is fully inside, and choose the farmer's simultaneous door-shutting to be at the midpoint of that window. (b) Tabulate the four events A, B (nose reaches the exit door), F (entrance door shuts) and R (exit door shuts) in both frames. (c) In the runner's frame, show explicitly that neither door ever touches the pole.

– Solution

(a) The window. $\gamma = 5/3$, so the pole is contracted to $15/\gamma = 9$ m in the barn frame. At $ct = 0$ the tail is at $x = 0$ and the nose at $x = 9$. The nose reaches $x = 10$ after the pole advances 1 m, which takes $ct = 1/0.8 = 1.25$ m. So the pole is wholly inside for $0 < ct < 1.25$ m (that is, for 4.17 ns), and the farmer shuts both doors at $ct = 0.625$ m.

(b) The table. Transform with $ct' = \gamma(ct - \beta x)$, $x' = \gamma(x - \beta ct)$ and $\gamma = 5/3$, $\beta = 0.8$:

EVENT	ct (M)	x (M)	ct' (M)	x' (M)
A tail reaches entrance	0	0	0	0
B nose reaches exit	1.25	10	−11.25	15
F entrance door shuts	0.625	0	+1.0417	−0.8333
R exit door shuts	0.625	10	−12.2917	+15.8333

Read the fourth column. In the barn frame F and R are simultaneous. In the runner's frame R happens at $ct' = -12.29$ and F at $ct' = +1.04$ — the exit door shuts 13.33 m/c = 44.5 ns *earlier*, matching (2.2.56). Note also $x'_B = 15$ m, the pole's proper length ✓, and that in the runner's frame the nose reaches the exit at $ct' = -11.25$, i.e. well *before* the tail reaches the entrance at $ct' = 0$. The pole is sticking out of both ends at once, which is what a 15 m pole does in a 6 m barn.

(c) Nobody gets hit. In the runner's frame the pole is at rest occupying $0 \leq x' \leq 15$. Track the doors. The exit door is the worldline $x = 10$; substituting $ct = ct'/\gamma + \beta \cdot 10$ into $x' = \gamma(x - \beta ct)$ gives

$$x'_{\text{exit}}(ct') = \gamma(1 - \beta^2)10 - \beta ct' = 6 - 0.8 ct'.$$

It reaches the nose, $x' = 15$, when $6 - 0.8ct' = 15$, i.e. $ct' = -11.25$ ✓ (event B). It shuts at $ct' = -12.29$, which is *earlier* — at that moment it sits at $x' = 6 + 0.8(12.2917) = 15.83$ m, which is beyond the nose. The door shuts in empty space in front of the pole, reopens, and the pole then passes through.

The entrance door is $x = 0$, worldline $x'_{\text{ent}}(ct') = -0.8 ct'$. It shuts at $ct' = +1.0417$, when it sits at $x' = -0.833$ m — behind the tail, which is at $x' = 0$. It too shuts in empty space.

The point. Every frame-independent question — did a door hit the pole? did the nose pass the exit before or after the exit door shut? — gets the same answer in both frames. Only "were the doors shut simultaneously?" differs, and that was never a fact about the world. If you insist on a barn that genuinely traps the pole, you must shut the doors and keep them shut, and then the runner's frame will report that the nose crumples against the closed exit door while the tail is still outside — a real, invariant, and rather expensive event that both frames agree on.

PROBLEM 3 — TWO BOOSTS IN DIFFERENT DIRECTIONS LEAVE A ROTATION BEHIND

Work in the (ct, x, y) subspace. (a) Show that a pure boost matrix — in any direction — is symmetric. (b) Compute $M = \Lambda_y(\phi_2)\Lambda_x(\phi_1)$ explicitly and show it is not symmetric unless one rapidity vanishes. Conclude that it cannot be a pure boost, so boosts do *not* form a group. (c) Show that the leftover is a rotation, and estimate its size for small rapidities. Quote (▮) the exact angle and check your estimate against it.

– Solution

(a) Pure boosts are symmetric. A boost along x is (2.2.48) padded with identity rows, manifestly symmetric. A boost in a general direction $\hat{n}$ is that one conjugated by a rotation R that takes $\hat{n}$ to $\hat{x}$: $\Lambda_{\hat{n}} = R^\top \Lambda_x R$ (with R acting trivially on ct). Then $\Lambda_{\hat{n}}^\top = R^\top \Lambda_x^\top R = R^\top \Lambda_x R = \Lambda_{\hat{n}}$ ✓. So symmetry is the algebraic signature of "pure boost, no rotation".

(b) The product. With $c_i = \cosh \phi_i$, $s_i = \sinh \phi_i$ and rows/columns ordered (ct, x, y) :

$$\Lambda_x(\phi_1) = \begin{pmatrix} c_1 & -s_1 & 0 \\ -s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \Lambda_y(\phi_2) = \begin{pmatrix} c_2 & 0 & -s_2 \\ 0 & 1 & 0 \\ -s_2 & 0 & c_2 \end{pmatrix},$$

$$M = \Lambda_y(\phi_2) \Lambda_x(\phi_1) = \begin{pmatrix} c_1 c_2 & -s_1 c_2 & -s_2 \\ -s_1 & c_1 & 0 \\ -s_2 c_1 & s_1 s_2 & c_2 \end{pmatrix}.$$

Compare entries across the diagonal: $M_{01} = -s_1 c_2$ against $M_{10} = -s_1$; these agree only if $c_2 = 1$, i.e. $\phi_2 = 0$. Likewise $M_{12} = 0$ against $M_{21} = s_1 s_2$. So M is not symmetric, hence by (a) not a pure boost.

Two boosts in different directions do not compose to a boost. The set of pure boosts is therefore not closed, and is not a group — only the boosts *along a fixed axis* are, which is what §3 actually proved.

(c) What is left over. ▮ Every proper orthochronous Lorentz transformation factors uniquely as $M = B R$ with B a pure boost and R a rotation (the polar decomposition; we quote it). Since M is not symmetric, $R \neq I$: there is a residual rotation, the **Wigner rotation**, in the xy plane.

Its size for small rapidities follows from non-commutativity. Write $\Lambda_x(\phi) = \exp(\phi K_x)$ and $\Lambda_y(\phi) = \exp(\phi K_y)$ with generators

$$K_x = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad K_y = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}.$$

Multiplying out, $[K_y, K_x] = K_y K_x - K_x K_y$ has entries only in the xy block:

$$[K_y, K_x] = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \equiv J_z,$$

the generator of rotations about z . ▮ By the Baker–Campbell–Hausdorff formula (Chapter 6.1), $e^A e^B = \exp\left(A + B + \frac{1}{2}[A, B] + \cdots\right)$, so

$$M = \exp\left(\phi_2 K_y + \phi_1 K_x + \frac{1}{2}\phi_1\phi_2 J_z + \cdots\right),$$

giving a rotation angle $\Omega \approx \frac{1}{2}\phi_1\phi_2 \approx \frac{1}{2}\beta_1\beta_2$ at low speed. It is *second order* in the velocities, which is why it has no Newtonian counterpart and why nobody stumbled on it before 1926.

▮ **The exact angle**, quoted, for two perpendicular boosts:

$$\tan \Omega = \frac{\gamma_1 \gamma_2 \beta_1 \beta_2}{\gamma_1 + \gamma_2}.$$

Check the low-speed limit: $\gamma_i \rightarrow 1$ and $\tan \Omega \rightarrow \Omega$ give $\Omega \rightarrow \beta_1\beta_2/2$ ✓, matching the commutator estimate. Numerically, for $\beta_1 = \beta_2 = 0.6$ the exact formula gives $\Omega = 0.2213$ rad = 12.7° , and the two leading-order estimates bracket it — $\frac{1}{2}\beta_1\beta_2 = 0.1800$ from below and $\frac{1}{2}\phi_1\phi_2 = 0.2402$ from above, since $\beta < \phi$ for $\beta > 0$. They are the same estimate to leading order and only differ at the next one: for $\beta_1 = \beta_2 = 0.01$ the exact angle is 5.00025×10^{-5} , against 5.00000×10^{-5} and 5.00033×10^{-5} — agreement to five figures.

Why it matters. An electron in an atom is continuously boosted in changing directions by the nuclear Coulomb field. The accumulated Wigner rotations make its spin precess — the **Thomas precession** — and this contributes a factor of $\frac{1}{2}$ to the spin–orbit coupling in atomic fine structure. Without it, the predicted fine-structure splitting is wrong by a factor of two, and the discrepancy stood unexplained for two years. A pure kinematic effect, with no force involved at all, showing up in a spectrometer.

PROBLEM 4 — A ROCKET AT CONSTANT PROPER ACCELERATION

A rocket accelerates so that its crew feel a constant a — that is, its acceleration measured in its own instantaneous rest frame is always a . (a) Show that its rapidity grows linearly in proper time, $\phi = a\tau/c$, and hence $\beta = \tanh(a\tau/c)$. (b) Find the lab-frame time and distance as functions of τ . (c) With $a = g = 9.81 \text{ m s}^{-2}$, compute β , the lab time and the distance after 1 year of proper time, and then the proper time needed to reach the galactic centre, 26 000 light-years away. Comment.

– Solution

(a) The rapidity grows linearly. This is where §6 pays for itself. In a proper-time interval $d\tau$, the rocket's instantaneous rest frame sees it acquire a velocity $dv = a d\tau$, hence a rapidity increment

$$d\phi = \text{artanh}\left(\frac{a d\tau}{c}\right) = \frac{a d\tau}{c} + O(d\tau^3).$$

Because *rapidities add* ((2.2.50)), these increments simply accumulate — no composition formula is needed, which is exactly what would have made the velocity version painful. Integrating from rest,

$$\phi(\tau) = \frac{a\tau}{c}, \quad \beta(\tau) = \tanh\frac{a\tau}{c}, \quad \gamma(\tau) = \cosh\frac{a\tau}{c}.$$

(b) Lab time and distance. By (2.2.31), $dt = \gamma d\tau = \cosh(a\tau/c) d\tau$, so

$$t(\tau) = \frac{c}{a} \sinh\frac{a\tau}{c}.$$

And $dx = v dt = c \tanh(a\tau/c) \cdot \cosh(a\tau/c) d\tau = c \sinh(a\tau/c) d\tau$, so

$$x(\tau) = \frac{c^2}{a} \left[\cosh\frac{a\tau}{c} - 1 \right].$$

Eliminating τ with $\cosh^2 - \sinh^2 = 1$ gives $(x + c^2/a)^2 - c^2 t^2 = (c^2/a)^2$: the worldline is a **hyperbola**. Chapter 2.3 will show that this is the spacetime analogue of a circle — the curve of constant "curvature" — and Chapter 3.1 will note that its crew experience something remarkably like a gravitational field, complete with a horizon behind them.

(c) Numbers. The natural scale is

$$\frac{c}{g} = \frac{2.998 \times 10^8}{9.81} = 3.06 \times 10^7 \text{ s} = 0.969 \text{ yr}, \quad \frac{c^2}{g} = 0.969 \text{ ly}.$$

A pleasant accident of our units: one year at one gee is very nearly one rapidity unit.

PROPER TIME τ	$\phi = g\tau/c$	β	γ	LAB TIME t	DISTANCE x
1 yr	1.032	0.7748	1.582	1.187 yr	0.564 ly
2 yr	2.065	0.9683	4.005	3.756 yr	2.911 ly
5 yr	5.161	0.999934	87.2	84.5 yr	83.5 ly
10 yr	10.32	0.9999999980	1.52×10^4	1.47×10^4 yr	1.47×10^4 ly

For the galactic centre, invert the distance formula: $\cosh \phi = 1 + xa/c^2 = 1 + 26\,000/0.969 = 2.684 \times 10^4$, so $\phi = 10.89$ and

$$\tau = \frac{c\phi}{g} = 0.9687 \times 10.891 = 10.55 \text{ years of crew time.}$$

(Lab time: 26 001 years. The Galaxy does not get a discount.)

Comment. Two things are worth extracting. First, the crew never detect anything odd: their accelerometer reads a constant g forever, their rapidity climbs without limit at 1.03 per year, and at no point do they approach a barrier. It is only the *velocity* that saturates, and velocity is the badly chosen coordinate — this is (2.2.46) made visceral. Second, the crew-time cost of a journey grows only *logarithmically* with distance once you are relativistic, since $\phi \approx \ln(2ax/c^2)$ for large x : the Andromeda galaxy, 2.5 million light-years away, costs 14.97 years of crew time against 10.55 for the galactic centre. The Universe is traversable within a lifetime and simultaneously unreachable, because you can never come home. Chapter 2.5 supplies the reason nobody does it: the fuel.

THE BRICK YOU JUST LAID

You derived (2.2.19) — $ct' = \gamma(ct - \beta x)$, $x' = \gamma(x - \beta ct)$, $y' = y$, $z' = z$ — from the two postulates plus homogeneity, isotropy and reciprocity, with every step forced and every assumption named. You showed it reduces to Galileo as $c \rightarrow \infty$, that boosts along an axis compose into boosts, that the composition is addition *in rapidity* and not in velocity, and that the quantity $c^2 t^2 - x^2 - y^2 - z^2$ is invariant even though you only ever demanded it for light. Simultaneity, dilation and contraction came out as corollaries in that order, and the two classical paradoxes turned out to be one paradox with two costumes.

Where this gets spent. Chapter 2.3 takes the invariant interval (2.2.21) and the hyperbolic form (2.2.48) and shows that together they define a *geometry* — spacetime, in which a boost is a rotation and proper time is arc length. Chapter 2.4 takes the matrix (2.2.24), calls it $\Lambda^\mu{}_\nu$, and defines a tensor as anything that transforms with it — the rapidity form and the group property of §3 and §6 are quoted there verbatim. Chapter 2.5 applies the same transformation to momentum and energy instead of position and time, and $E = mc^2$ falls out of it; the γ you derived here is the γ in $E = \gamma mc^2$. Chapter 2.6 shows that the electric and magnetic fields transform into each other under exactly this boost, which retroactively explains Chapter 2.1's entire crisis. Chapter 3.1 keeps the transformation but only *locally*, which is the whole idea of general relativity. And Chapter 6.1 returns to the rapidity ϕ and names what it has been all along: the parameter of a one-parameter Lie group, with the boost generator K as its Lie-algebra element.